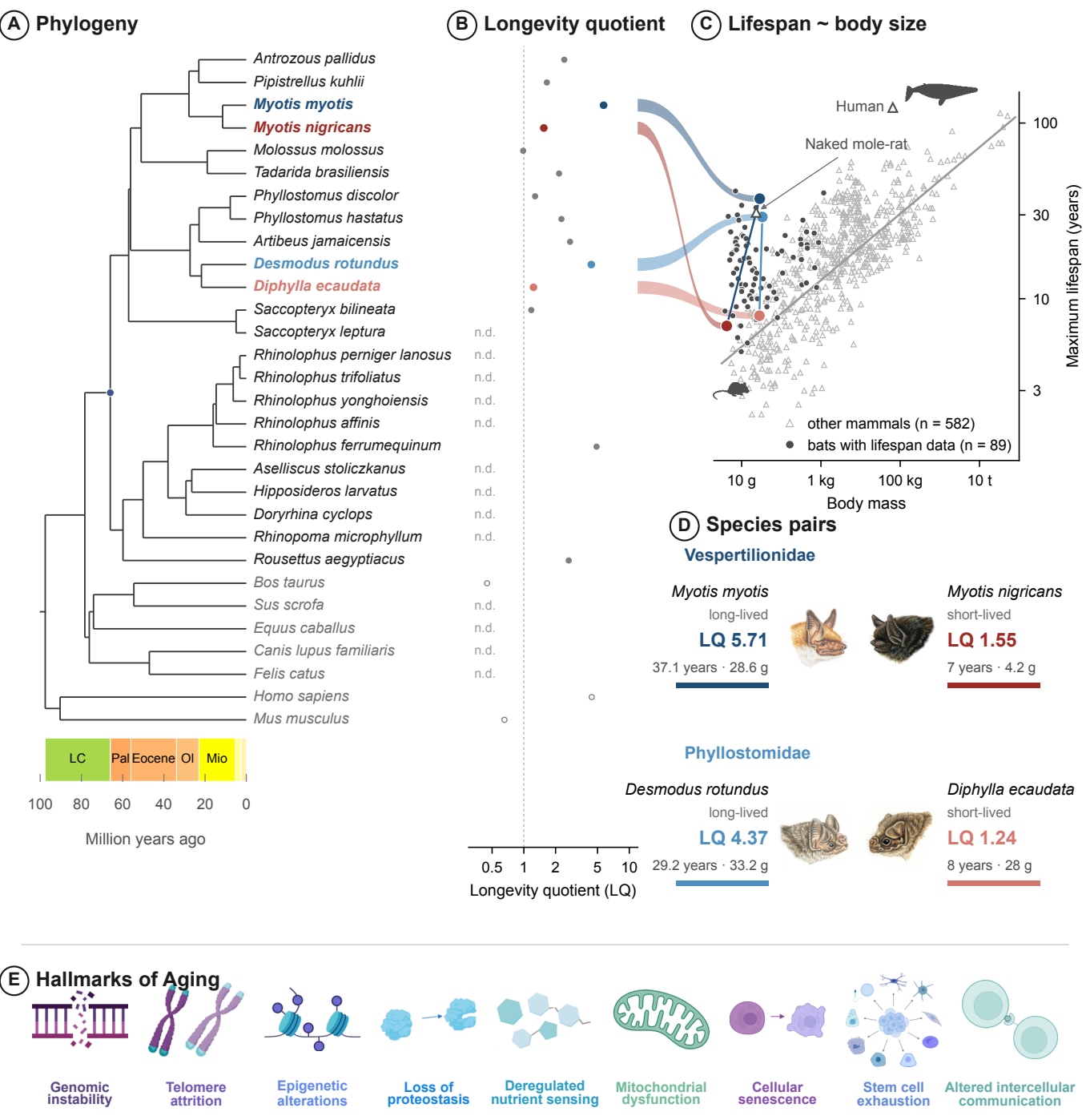

A Phylogeny
Antrozous pallidus
Pipistrellus kuhlii
Myotis myotis
Myotis nigricans
Molossus molossus
Tadarida brasiliensis
Phyllostomus discolor
Phyllostomus hastatus
Artibeus jamaicensis
Desmodus rotundus
Diphylla ecaudata
Saccopteryx bilineata
Saccopteryx leptura
Rhinolophus perniger lanosus
Rhinolophus trifoliatus
Rhinolophus yonghoiensis
Rhinolophus affinis
Rhinolophus ferrumequinum
Aselliscus stoliczkanus
Hipposideros larvatus
Doryrhina cyclops
Rhinopoma microphyllum
Rousettus aegyptiacus
Bos taurus
Sus scrofa
Equus caballus
Canis lupus familiaris
Felis catus
Homo sapiens
Mus musculus
n.d.
LC
Pal
Eocene
Ol
Mio
100 80 60 40 20 0
Million years ago
B Longevity quotient
0.5 1 2 5 10
Longevity quotient (LQ)
C Lifespan ~ body size
Human
Naked mole-rat
other mammals (n = 582)
bats with lifespan data (n = 89)
Maximum lifespan (years)
100
30
10
3
10 g
1 kg
100 kg
10 t
Body mass
D Species pairs
Vespertilionidae
Myotis myotis
long-lived
LQ 5.71
37.1 years · 28.6 g
Myotis nigricans
short-lived
LQ 1.55
7 years · 4.2 g
Phyllostomidae
Desmodus rotundus
long-lived
LQ 4.37
29.2 years · 33.2 g
Diphylla ecaudata
short-lived
LQ 1.24
8 years · 28 g
E Hallmarks of Aging
Genomic instability
Telomere attrition
Epigenetic alterations
Loss of proteostasis
Deregulated nutrient sensing
Mitochondrial dysfunction
Cellular senescence
Stem cell exhaustion
Altered intercellular communication